

Estimation of Change in Population of municipalities in PR

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Presentation Overview

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- Demographic change is an important topic for governments and policymakers.
- Understanding population characteristics is particularly important for regional governments.
- Anticipating population change helps guide infrastructure and public investment decisions.

- Data obtained from the American Community Survey (ACS) 5-year estimates U.S. Census Bureau
- Population estimates from 2011 to 2023 were used.
- Dataset includes all 78 municipalities in Puerto Rico.

Results

Densities

Credibility interval

Confidance Interval

Distribution

- Results did not show statistically significant changes over time.
- Due to this lack of significance, strong predictive conclusions cannot be made.
- No meaningful recommendations can be derived from the current model.

Conclusions

- Simple regression results were inconclusive.
- Population data are highly correlated with previous years.
- This violates the i.i.d. assumption of simple regression.
- Time series models may be more appropriate for this problem.